

DIP HW3_1

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1. Please depict the histogram and graph of the assigned image “aerial_view.tif”, and print out the source code? (10)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. 讀取影像 (灰階)
image = cv2.imread("aerial_view.tif", cv2.IMREAD_GRAYSCALE)

# 2. 計算直方圖
hist = cv2.calcHist([image], [0], None, [256], [0, 256])
hist_normalized = hist / hist.sum()

# 3. 同一張 figure 顯示原圖 & 直方圖
plt.figure(figsize=(10,4))

# 原始影像
plt.subplot(1,2,1)
plt.imshow(image, cmap="gray")
plt.title("Original Image")
plt.axis("off")

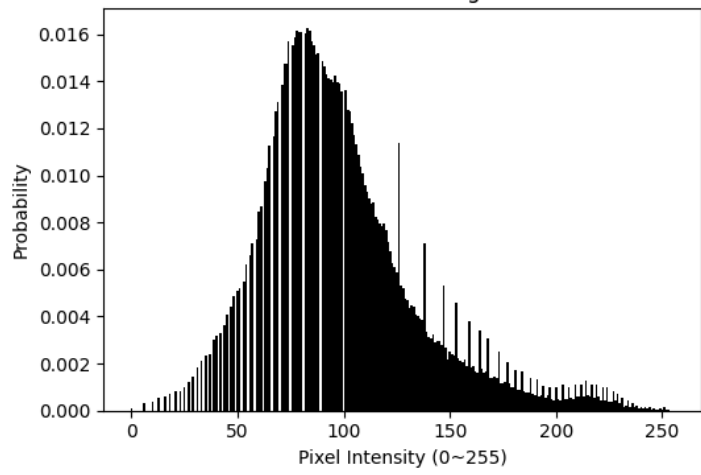
# 直方圖
plt.subplot(1,2,2)
plt.bar(range(256), hist_normalized.flatten(), width=1.0,
color='black')
plt.title("Normalized Histogram")
plt.xlabel("Pixel Intensity (0~255)")
plt.ylabel("Probability")

plt.tight_layout()
plt.show()
```

Original Image



Normalized Histogram



2. Please plot the histogram and graph of the image after Histogram Equalization, and print out the source code? (30)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. 讀取影像 (灰階)
image = cv2.imread("aerial_view.tif", cv2.IMREAD_GRAYSCALE)

# 2. 直方圖均衡化
equalized_image = cv2.equalizeHist(image)

# 3. 計算直方圖
hist_eq = cv2.calcHist([equalized_image], [0], None, [256], [0, 256])
hist_eq_normalized = hist_eq / hist_eq.sum()

# 4. 顯示影像 + 直方圖 (同一張圖)
plt.figure(figsize=(12,5))

# (左邊) 均衡化影像
plt.subplot(1,2,1)
plt.imshow(equalized_image, cmap="gray")
plt.title("Image After Equalization")
plt.axis("off")

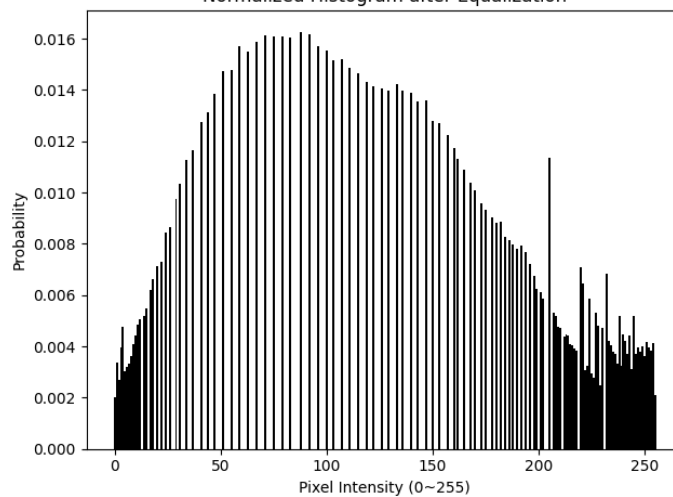
# (右邊) 直方圖
plt.subplot(1,2,2)
plt.bar(range(256), hist_eq_normalized.flatten(), width=1.0,
color='black')
plt.title("Normalized Histogram after Equalization")
plt.xlabel("Pixel Intensity (0~255)")
plt.ylabel("Probability")

plt.tight_layout()
plt.show()
```

Image After Equalization



Normalized Histogram after Equalization



3. Please plot the histogram and graph of the image after Histogram

Matching (specification) by $p_z(z_q) = c \cdot z_q^{0.4}$, and print out the source code?

(NOTE: the parameter, c , needs to calculate in advance) (40)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. 讀取影像 (灰階)
image = cv2.imread("aerial_view.tif", cv2.IMREAD_GRAYSCALE)

# 2. 計算常數 c
z = np.arange(256)
c = 1.0 / np.sum(z**0.4)
target_pdf = c * (z**0.4)
target_pdf[0] = 0 # 避免 z=0 出現 nan
target_pdf = target_pdf / target_pdf.sum() # 正規化

# 3. 計算目標 CDF
target_cdf = np.cumsum(target_pdf)

# 4. 計算原始影像的直方圖 & CDF
orig_hist = cv2.calcHist([image], [0], None, [256], [0, 256]).flatten()
orig_pdf = orig_hist / orig_hist.sum()
orig_cdf = np.cumsum(orig_pdf)

# 5. 建立映射 (原始灰階 → 目標灰階)
mapping = np.zeros(256, dtype=np.uint8)
for i in range(256):
    diff = np.abs(target_cdf - orig_cdf[i])
    mapping[i] = np.argmin(diff)

# 6. 應用映射, 生成新影像
matched_image = mapping[image]

# 7. 計算匹配後直方圖
matched_hist = cv2.calcHist([matched_image], [0], None, [256], [0, 256])
matched_hist = matched_hist / matched_hist.sum()

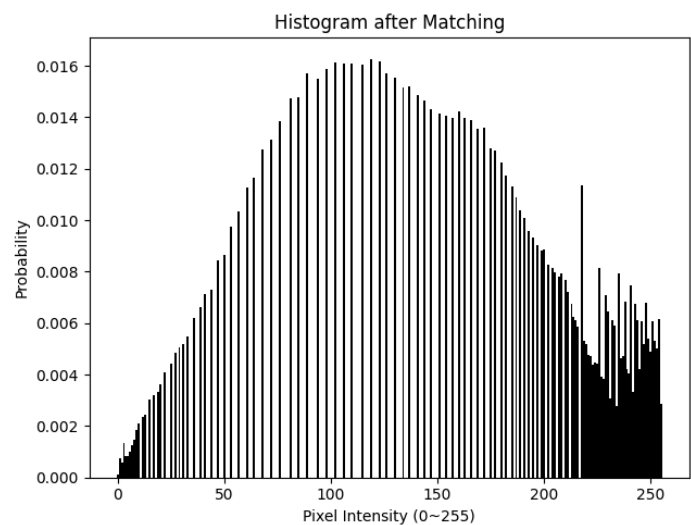
# 8. 顯示影像與直方圖 (並排)
```

```
plt.figure(figsize=(12,5))

# (左) 規格化後影像
plt.subplot(1,2,1)
plt.imshow(matched_image, cmap="gray")
plt.title("Histogram Matched Image")
plt.axis("off")

# (右) 規格化後直方圖
plt.subplot(1,2,2)
plt.bar(range(256), matched_hist.flatten(), width=1.0, color='black')
plt.title("Histogram after Matching")
plt.xlabel("Pixel Intensity (0~255)")
plt.ylabel("Probability")

plt.tight_layout()
plt.show()
```



4. Please comment the original, the histogram-equalized and the histogram-matching images ? (20)

在原始影像中，可以觀察到亮度分布並不均勻，直方圖集中於某些範圍，導致整體的對比度有限。由於灰階值分布不夠廣泛，影像的明暗差異不明顯，使得部分細節容易被忽略，整體感覺偏灰暗或偏亮。

經過直方圖均衡化處理後，影像的灰階值被重新分布，使得直方圖在 0 到 255 的範圍內更為平均。這樣的結果顯著提升了影像的對比度，暗部細節變得更加清晰，亮部也更容易區分，整體畫面呈現出更強烈的層次感。不過，在某些情況下，均衡化可能會造成部分區域過度增強，使影像看起來略顯不自然。

至於直方圖規格化，則是將影像的直方圖調整為符合指定的機率分布 $p_z(z_q) = c \cdot z_q^{0.4}$ 。在這樣的處理下，影像的灰階值分布不會像均衡化那樣完全平均，而是依照給定的函數趨勢進行調整。結果顯示，規格化後的影像在亮度與對比度的表現上較為柔和，既能提升細節，又能避免過度增強，因此視覺效果比均衡化更自然。同時，它也能呈現特定的影像風格，適合在需要模擬特定亮度分布或影像效果的情境中使用。